Machine Learning for Spectral Analysis

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1. Introduction

Optical Emission Spectroscopy (OES) provides **non-invasive, real-time diagnostics** of plasma processes by analysing emission spectra from excited species. Plasma-based manufacturing (semiconductor etching, surface treatment, thin-film deposition) relies on OES for process monitoring, but manual spectral interpretation is slow and error-prone.

This project develops an **automated ML pipeline** for plasma OES analysis targeting four objectives:

- **1. Feature ID** — Peak detection, NMF, NIST matching
- **2. Classification** — SVM, RF, CNN, Transformer (6 models)
- **3. Temporal** — Attention-LSTM, Species time-series
- **4. Intensity** — Actinometry, Boltzmann Te

Three public datasets:

Dataset	Channels	Notes
LIBS Benchmark	40,002 ch	Steel composition (12 classes)
Mesbah CAP	51 ch	N2 plasma T_rot / T_vib
BOSCH RIE	3,648 ch	25 Hz, 10 days, SF6/C4F8

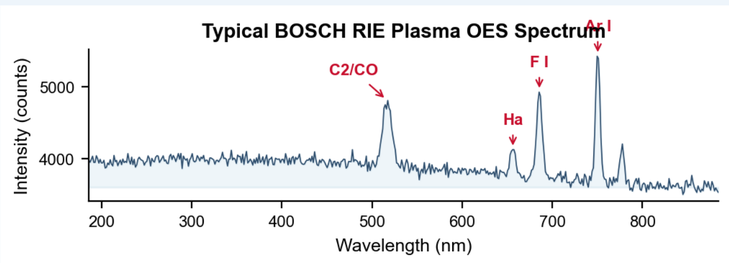

Fig. 1: Simulated BOSCH RIE plasma OES spectrum with annotated species

2. Methodology

Raw OES Spectra (3,648 channels)

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Preprocessing
Cosmic Ray → ALS Baseline → SavGol Smooth → SNV Normalise

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Feature Extraction
PCA | NIST Line Selection | NMF Decomposition | PlasmaDescriptor

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Model Training (6 architectures)
PLS | Ridge/SVM/RF | CNN | LSTM | Attention-LSTM | Transformer

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Evaluation & Interpretability
5-fold CV | SHAP | Boltzmann Plot | Actinometry

Hyperparameter optimisation: Optuna two-stage search (20 trials per target)

Key Design Decisions

Decision	Rationale
Per-element routing	Cr: Ridge+PCA; others: ANN+NIST
GroupKFold CV	Prevents same-target leakage
Balanced weights	OFF=12.1%, weighted CE loss
NMF over PCA	Non-negative = physical spectra

Preprocessing rationale: ALS baseline ($\lambda=10^5$) removes fluorescence continuum. Savitzky-Golay (window=11, order=3) preserves peak shapes while reducing shot noise. SNV normalisation corrects for optical path length variation between measurements. Cosmic ray removal uses Z-score median filter (threshold=5σ, 11-channel local window).

3. Species Identification

Non-negative Matrix Factorization (NMF) decomposes the spectral matrix **X** ≈ **W · H**, where each row of **H** is a pure-species emission spectrum and **W** contains the corresponding concentrations. NMF is physically appropriate because emission intensities are inherently non-negative.

Automated NIST matching: Each detected peak is compared against 39 reference emission lines from 13 species. The algorithm selects the closest database match within +/-1.5 nm tolerance. Species with peak intensity > μ + 3σ (global spectrum statistics) are classified as *present*.

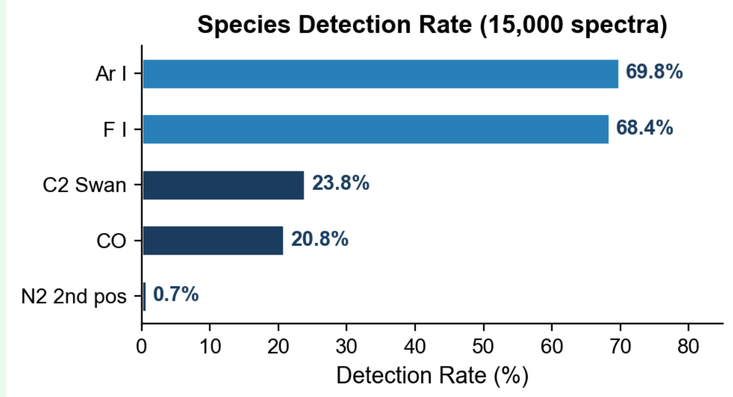

Fig. 2: Automated species detection rates across 15,000 OES spectra

Species	Lines (nm)	Origin
F I	685.6, 690.2, 703.7	SF6 etchant radical
Ar I	696.5, 750.4, 763.5	Carrier gas reference
C2 Swan	473.7, 516.5, 563.6	C4F8 carbon indicator
CO	451.1, 519.8, 561.0	C4F8 + O2 product
Si I	250.7, 252.4, 288.2	Si etch product
SiF	440.0, 442.5	Si + F recombination

Validation: NMF Component 0 peaks at 684.4 nm (≈ F I 685.6); Component 2 at 515.1 nm (approx. C2 Swan 516.5) — unsupervised decomposition confirms NIST species independently.

4. Classification Results

94.2% Accuracy (SVM/RF, 5-fold CV)

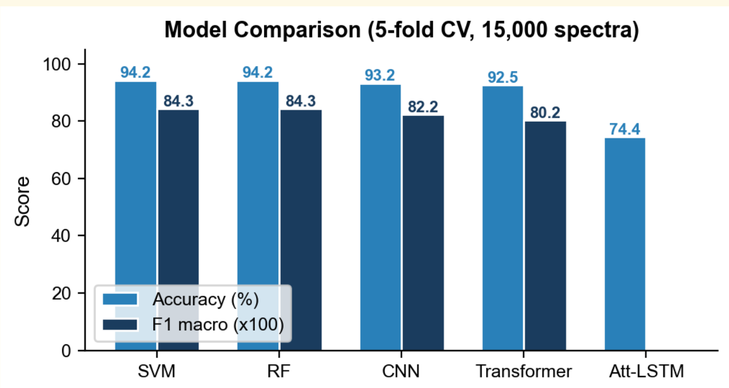

Fig. 3: 5-model accuracy and F1 comparison (5-fold CV)

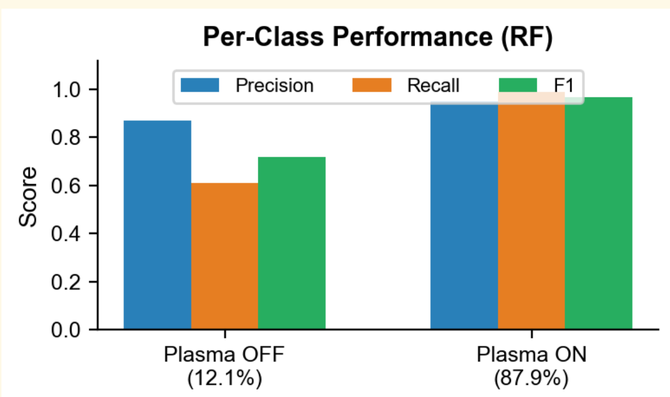

Fig. 4: Per-class precision, recall, and F1 score

- **ML > DL:** Near-linear decision boundary favours SVM/RF. DL needs more data to match.
- **Class imbalance:** OFF only 12.1%, recall=0.61. Balanced weights help but don't fully eliminate bias.
- **Physics validated:** Ar/F dominant (SF6/Ar gas); C2 only during C4F8 passivation steps.

5. Interpretability & Physics

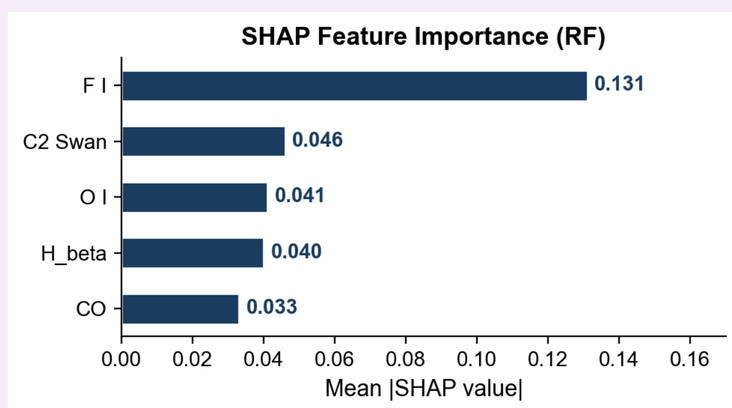

Fig. 5: SHAP feature importance — F I dominates (0.131)

Root Cause Discovery: Etch/passivation spectral difference < 0.35σ (indistinguishable). RF power ON/OFF provides 0.56σ separation. Classification: 74% → **94%.**

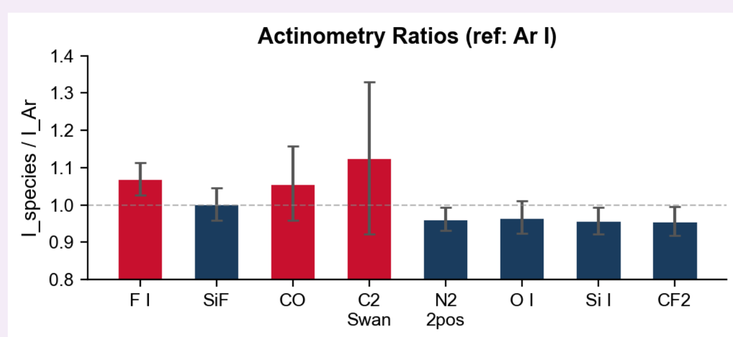

Fig. 6: Actinometry ratios (I_species / I_Ar, Coburn & Chen 1980)

Boltzmann T_exc = 13,334 K (6 Ar I lines). **Temporal:** Attention-LSTM 74.4% phase classification; DTW K-Means (k=4) discovers 4 discharge phases with F I 684 nm emission ratio **2.04x** between clusters.

6. Conclusions & Further Work

Key Achievements:

- ✓ **94.2%** plasma state classification (6 models compared)
- ✓ **13 species** auto-detected via NMF + NIST (39 lines)
- ✓ **SHAP:** F I = 0.131 importance (primary etchant, validated)
- ✓ **Label correction:** 74% → 94% via root-cause analysis
- ✓ **T_rot = 20.0 K, T_vib = 102.0 K** (CAP regression)
- ✓ **78 tests,** 6 CLI modes, fully reproducible

Limitations:

- OES→process parameter regression failed (R2 < 0, causal irreversibility)
- Boltzmann Te limited by narrow Ar I energy spread (0.33 eV)
- Spatial etch prediction lacks wafer ID alignment

Further Work:

- Extend to multi-class species classification (beyond binary ON/OFF)
- Implement real-time OES monitoring with streaming inference
- Apply transfer learning across different plasma reactors

Project Metrics: 15,000 spectra analysed | 3,648 spectral channels | 13 species identified | 39 NIST emission lines | 6 ML models compared | 78 automated tests | 32 development stories | 23 literature references | 6 CLI task modes | 3 public datasets

References:

[1] Gidon *et al.* (2019) IEEE Trans. Radiat. Plasma Med. Sci.
[2] Coburn & Chen (1980) J. Appl. Phys. — Actinometry
[3] Vaswani *et al.* (2017) NeurIPS — Transformer
[4] Contreras *et al.* (2024) Anal. Chem. — Spectral-zone SHAP
[5] BOSCH dataset: Zenodo #17122442